DATA TRANSMISSION FUNDAMENTALS

Packets vs. Frames

Understanding the architecture of network communication, from logical routing to physical delivery.

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1. THE BIG PICTURE

OSI Model Representation

Data Division

When you send data (like a message, video, or file), it does **NOT** travel as one big chunk. It is systematically divided into smaller units at different network layers.

Application: Data / Message

Transport: Segment (TCP) / Datagram (UDP)

Network Layer: Packet

Data Link Layer: Frame

2. WHAT IS A PACKET?

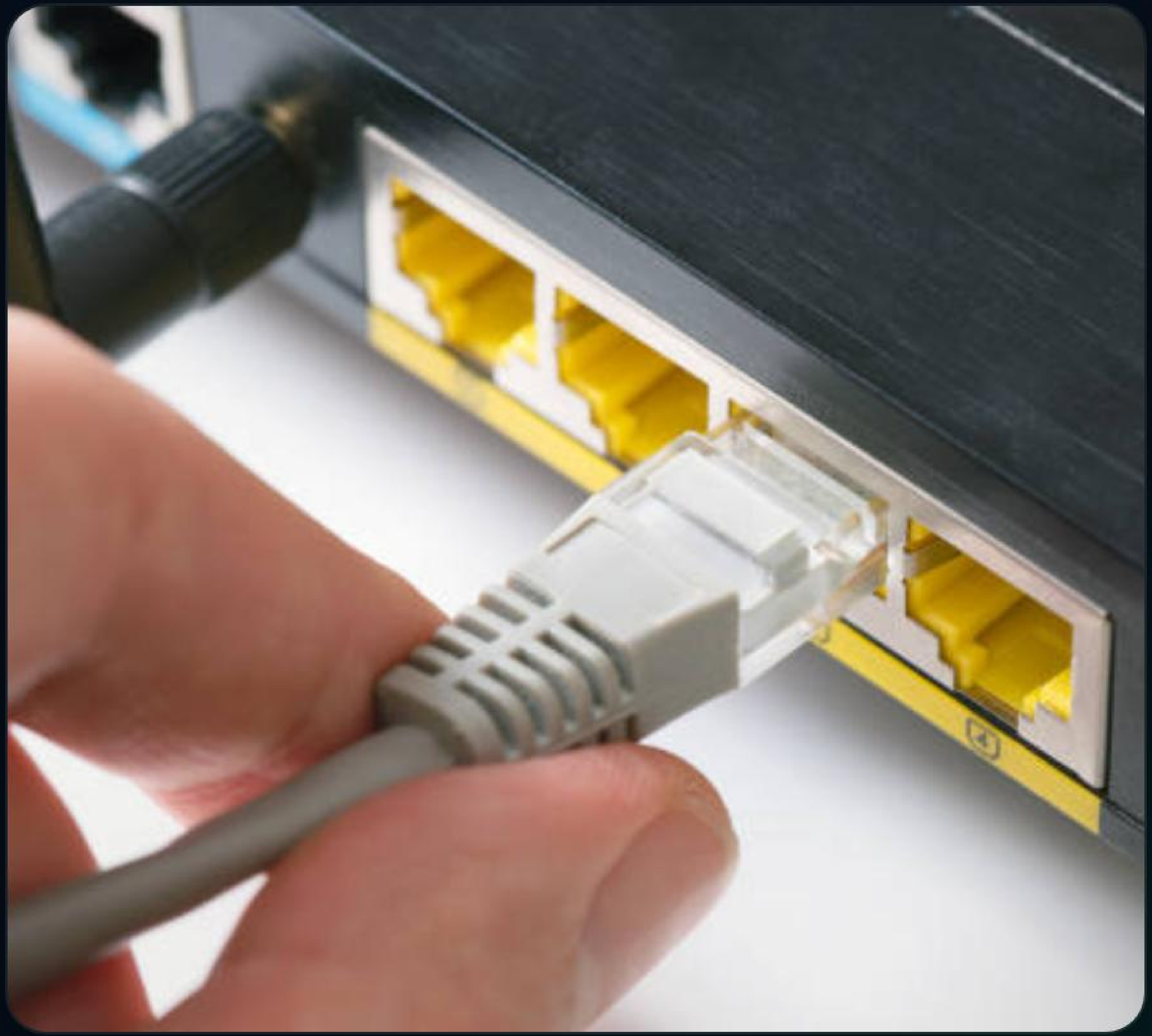
Definition: A small unit of data formatted for transmission at the **Network Layer (Layer 3)**.

Primary Purpose

- Routing data from source to destination across diverse networks.
- Providing logical addressing (IP Addressing).

Key Functions

- Logical IP Addressing
- Internet Routing
- Fragmentation
- Header Error Detection



PACKET STRUCTURE (IPV4)

A packet is constructed using a **Header** and the actual **Data (Payload)**.

Field	Size	Description
Version & Header Length	8 bits total	IP version (IPv4 or IPv6) & Size of header
Type of Service	8 bits	Priority handling instructions
Total Length	16 bits	Entire packet size limit
ID, Flags, Fragment Offset	32 bits total	Controls for data fragmentation
TTL (Time to Live)	8 bits	Prevents infinite looping within the network
Protocol & Header Checksum	24 bits total	Upper layer protocol identifier & Error detection
Source IP	32 bits	Sender's logical IP address
Destination IP	32 bits	Receiver's logical IP address

3. WHAT IS A FRAME?

Definition: A unit of data at the **Data Link Layer (Layer 2)** used for local network communication.

Primary Purpose

- Physical delivery (node-to-node on the same link).
- Utilizes MAC (Media Access Control) addresses.

Key Functions

- **Physical Addressing:** Using hardware MAC addresses.
- **Error Detection:** Full frame checking via CRC.
- **Flow Control:** Managing data pace between nodes.
- **Data Encapsulation:** Wrapping the IP Packet.

FRAME STRUCTURE (ETHERNET)

Unlike a packet, an Ethernet frame has a **Header + Data + Trailer**.

Field	Size	Description
Preamble	7 bytes	Bit synchronization for incoming data
Start Frame Delimiter	1 byte	Start indicator for the frame structure
Destination MAC	6 bytes	Receiver's physical hardware address
Source MAC	6 bytes	Sender's physical hardware address
Type / Length	2 bytes	Protocol type identifier (e.g., IPv4)
Data (Payload)	46–1500 bytes	The encapsulated Packet (IP data)
CRC (Trailer)	4 bytes	Cyclic Redundancy Check (Full error detection)

4. ENCAPSULATION PROCESS

How data is wrapped as it moves down the OSI model.

Application	Transport	Network	Data Link	Physical
Creates Data	Adds Header → **Segment**	Adds IP Header → **Packet**	Adds MAC + CRC → **Frame**	Converts to **Bits**

```
[ MAC Header | [ IP Header | [ TCP Header | DATA ] ] | CRC Trailer ]
```

5. PACKET VS. FRAME

Feature	Packet	Frame
Layer	Network Layer (Layer 3)	Data Link Layer (Layer 2)
Address Type	**IP Address** (Logical)	**MAC Address** (Physical)
Purpose	Routing between different networks	Delivery within a local physical network
Hardware Device	Router	Switch
Error Checking	Header checksum only	Full frame CRC check
Encapsulation	Contains a Transport Segment	Contains a Network Packet

6. REAL-LIFE EXAMPLE

Scenario: Sending a WhatsApp Message

You type "Hi" and press send. The data is instantly divided:

1. **Segment (TCP):** Created for reliable transport.
2. **Packet (IP):** Wraps the segment. The **Router** uses this packet (Destination IP) to navigate the internet.
3. **Frame (Ethernet/WiFi):** Wraps the packet. The local **Switch or Access Point** uses this frame (MAC Address) to deliver the data to the specific gateway or device.

7. FRAGMENTATION & 8. ERROR HANDLING

Fragmentation (Packet Level)

If a packet is too large for the network link (MTU exceeded):

- It is divided into smaller packets.
- Each packet fragment travels separately.
- Reassembled **only** at the final destination.

Key Fields Used: Identification, Fragment Offset, Flags.

Error Handling

Packet: Uses Header Checksum.

Detects errors in the IP header only (does not verify payload integrity).

Frame: Uses CRC (Cyclic Redundancy Check).

Appended as a trailer, it detects errors across the *entire* data frame.

9. DEVICES & 11. FRAME TYPES

Devices Involved

Device	Works On
Switch	Frames (Layer 2)
Router	Packets (Layer 3)
NIC (Network Card)	Frames (Layer 2)

Types of Frames

- **Data Frame:** Carries the actual payload data.
- **Control Frame:** Manages communication & flow control.
- **Management Frame:** Handles network setup and link control.

12. EXAM CONCEPTUAL QUESTION

”Why do we need both packets and frames?”

Packets handle **logical delivery** at the Internet level. They use IP addresses to find the global path to the destination network.

Frames handle **physical delivery** at the local network level. They use MAC addresses to move data hop-by-hop between physical devices (like from your PC to the router).

13. CORE SUMMARY

Packet

Logical delivery via **IP Addresses**. Managed by **Routers** to navigate between networks.

Frame

Physical delivery via **MAC Addresses**. Managed by **Switches** for local node delivery.

Encapsulation

The **Packet** is encapsulated *inside* the **Frame** payload for its physical journey.

Questions?

Thank you for your attention.

TEAM MEMBERS

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IMAGE SOURCES

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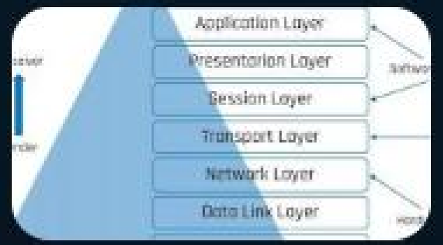

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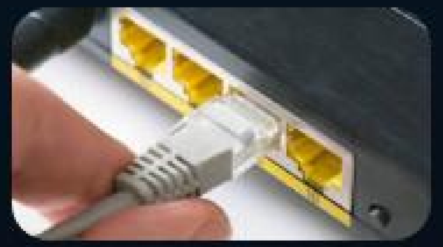

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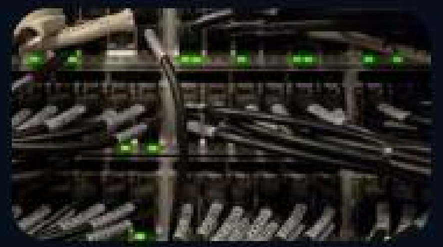

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